StyCLIP: StyleTransfer in Few-Shot Learning

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Introduction & Motivation

- **Context:** Rapid digitization of art collections demands automated artist classification systems for large-scale analysis.
- **Challenge:** Few-shot visual artist classification – identifying artist by their style with limited labeled examples per artist.
- **Problem:** Current image classifiers are based on semantic features, not style (e.g., CLIP).
- **Hypothesis:** Explicitly integrating style representations (Gram matrices) into CLIP could enhance artist-specific pattern recognition.
- **Contribution:** StyCLIP – a CLIP extension incorporating dedicated style-aware features for improved few-shot artist classification.

Related Work

Style Transfer

- Gatys et al. [3]: Deep networks encode style in intermediate layers; Gram matrices capture style.

Few-Shot Learning (FSL) & Meta-Learning

- Prototypical Networks [6]: Learn to learn from small samples.

Contrastive Language-Image Pretraining (CLIP)

- CLIP [5]: Shared embedding space for images and text.
- TIP-Adapter [8]: Non-parametric cache model for FSL.
- APE [9]: Adaptive prior refinement of CLIP features.
- GDA-CLIP [7]: Gradient-based adaptation.
- TIMO [4]: Mutual guidance between text/image modalities.

Dataset

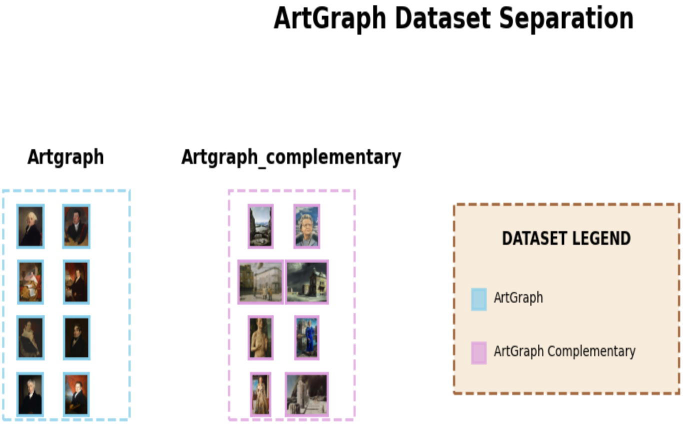
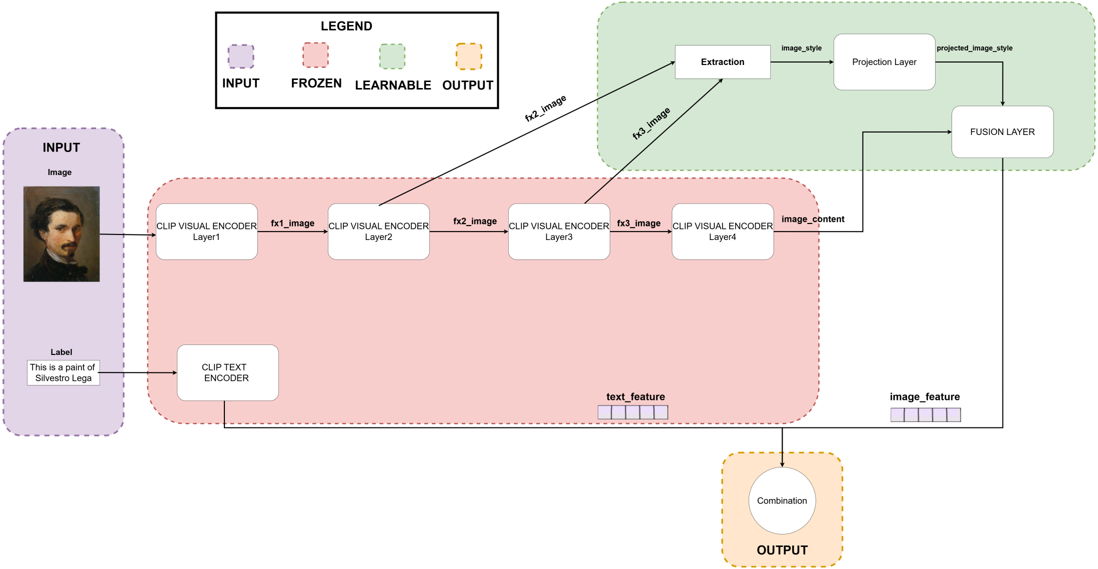


Figure: ArtGraph Datasets Overview

StyCLIP: Architecture Overview



StyCLIP: Key Components

- **Core Idea:** Extend CLIP with explicit style-aware features via Gram matrices to capture artist-specific patterns
- **Three-Stage Architecture:**
  1. **Gram Matrix Extraction:** Extract style correlations from ResNet-50 intermediate layers

     $$G_{i,j}^{(l)} = \frac{1}{H \times W} \sum_{k=1}^{H \times W} F_{i,k}^{(l)} \cdot F_{j,k}^{(l)}$$

     - $F^{(l)} \in \mathbb{R}^{C \times H \times W}$: feature maps from layer l (reshaped to $C \times HW$)
     - $G_{i,j}^{(l)}$: correlation between channels i and j (captures style patterns)
     - Normalization by $H \times W$ ensures scale-invariance across image sizes
  2. **Style Projection:** Compress high-dimensional Gram matrices into compact representations

     $$\mathbf{s} = \mathrm{MLP}(\mathrm{concat}[\mathrm{vec}(\mathrm{triu}(G^{(2)})), \mathrm{vec}(\mathrm{triu}(G^{(3)}))])$$
  3. **Adaptive Fusion:** Integrate semantic and style features via bottleneck adapter

     $$\mathbf{f}_{\text{fused}} = \mathrm{Adapter}([\mathbf{F}_{\text{v}}; \mathbf{s}])$$

StyCLIP: Training Strategy

- **Frozen Backbone:** Original CLIP (RN50) visual and text encoders are frozen.
- **Trainable Components:** Only the new style projection layers and fusion adapter are optimized.
- **Rationale:**
  - Preserve pre-trained semantic knowledge from CLIP.
  - Reduce computational cost.
  - Enable efficient fine-tuning on artistic datasets.

Hyperparameter Tuning

- Optuna [1] used for:
  - Fusion bottleneck dimension {64, 128, 256}
  - Dropout rate [0.0, 0.5]
  - Learning rate log-uniform $[10^{-5}, 10^{-3}]$
  - Weight decay log-uniform $[10^{-6}, 10^{-3}]$

Experimental Setup

Meta-Learning Training for StyCLIP

- Prototypical network approach.
- Episodic training: 10-way (N=10 artists), K-shot (K=1, 4) classification, 3-query (Q=3)

Evaluation Protocol

- **Method:** Using every CLIP-based method (e.g., TIMO, TIP-Adapter) with every backbone.
- N-way classification (all available artists in the test subset).
- Tested across K $\in \{1, 2, 4, 8, 16\}$ shots for support set.
- **Statistical Robustness:** 10 different random seeds.
- **Metrics:** Accuracy, Macro-Precision, Macro-Recall, Macro-F1.

Episode Example for FSL

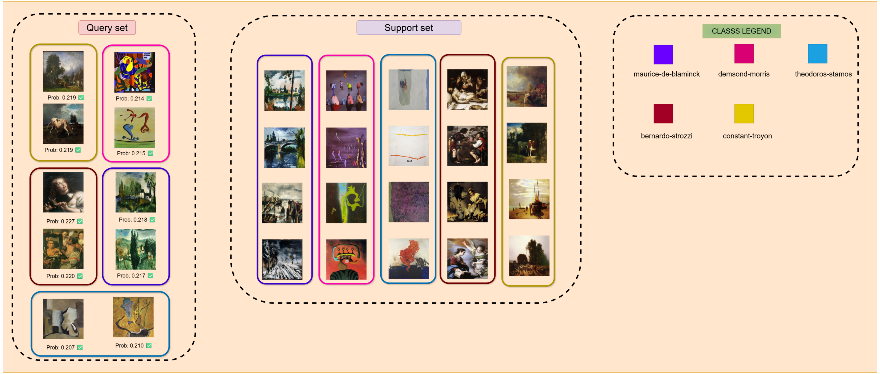


Figure: Example episode: 5-Way, 4-Shots ,10 query examples

Key Evaluation Results

Main Finding: Integrating explicit style-aware features (StyCLIP) *consistently underperforms* standard CLIP variants.

- Standard CLIP with RN50 backbone is the best performing backbone.
- StyCLIP backbones (StyCLIP_FSL1, StyCLIP_FSL4) significantly degrade performance for all FSL methods.
- TIMO and TIMO-S are the best FSL methods when using the standard RN50 backbone.

Example: 16-Shot Classification Accuracy (%)

Model	Backbone	ACC (%)
TIMO	StyCLIP_FSL1	37.00
TIMO	StyCLIP_FSL4	43.27
TIMO	**RN50**	**73.36**
TIMO_S	StyCLIP_FSL1	39.20
TIMO_S	StyCLIP_FSL4	48.75
TIMO_S	**RN50**	**73.46**

Statistical Analysis Summary

- T-tests confirmed the significance of performance differences.
- **Key Observation:** Standard CLIP (RN50) significantly outperforms StyCLIP-trained backbones across nearly all configurations and few-shot methods.
- Example: For 16-shot with TIMO_S:
  - RN50 (73.46%) vs StyCLIP_FSL4 (48.75%) $\rightarrow \approx$ 25 percentage point difference, statistically significant.

$\Longrightarrow$ The architectural changes in StyCLIP for explicit style modeling did not yield benefits and often hindered performance.

Conclusion & Discussion

- **Main Finding:** Integrating explicit style-aware features via Gram matrices (StyCLIP) did **not improve** few-shot artist classification performance.
- **Performance:** Standard CLIP (RN50 backbone) consistently outperformed StyCLIP variants.
- **Implication:** Explicit Gram matrix-based style modeling might interfere with CLIP's pre-trained semantic representations rather than enhance them for this task.
- **Future Directions:** Alternative approaches like attention-based style integration or domain-specific fine-tuning (without architectural changes) might be more fruitful.

Acknowledgments

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Code Availability

The code for this project is available on GitHub:

`https://github.com/Fonty02/ComputerVision/tree/main/TIMO`

Thank You! 🚀

Appendix: Hyperparameter Optimization Highlights

StyCLIP FSL1 (1-shot trained)

- Bottleneck: 64
- LR: 6.01×10^{-5}
- Dropout: 0.0
- Weight Decay: 4.83×10^{-4}
- Valid Acc: 57.07%

StyCLIP FSL4 (4-shot trained)

- Bottleneck: 128
- LR: 1.02×10^{-5}
- Dropout: 0.1
- Weight Decay: 1.03×10^{-6}
- Valid Acc: 60.53%

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